Technology & Mental Health

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Motivation

Technology is everywhere, and when not used responsibly, people think it can have a negative impact. Individuals already have enough to manage in their daily lives, and in today’s world, technology adds yet another layer of complexity — from too much television to gaming on computers and consoles to social media and the countless digital platforms we interact with every day.

Because of this, I thought it would be interesting to explore whether there is a relationship between technology usage and stress or overall mental health.

Data Sources:

1. **Dataset Name:** mental_health_and_technology_usage_2024

🏢 💻 Mental Health & Technology Usage Dataset 🌱

- Rows: 10,000
- Columns: 14
- Last Updated: 2024

2. **Dataset Name:** social_media_sleep_stress_productivity_11000

Social Media Usage vs Sleep, Stress & Productivity

- Rows: 11,000
- Columns: 10
- Last Updated : 2026

Methodology

Tools: Python, Jupyter Notebooks

Libraries: Pandas, Seaborn, Matplotlib,

Types of visualizations: Bar Plot, Box Plots, Correlation Heat Map

Data Preparation

- Two datasets were downloaded to (.csv) format.
- Files were imported to Jupyter Notebooks and merged into a single data frame.

Data Cleaning:

- After merging, reviewed the columns and identified 'duplicates' due to mismatch names.
- Renamed columns for alignment, considering case sensitivity.
- Remove columns with null values.

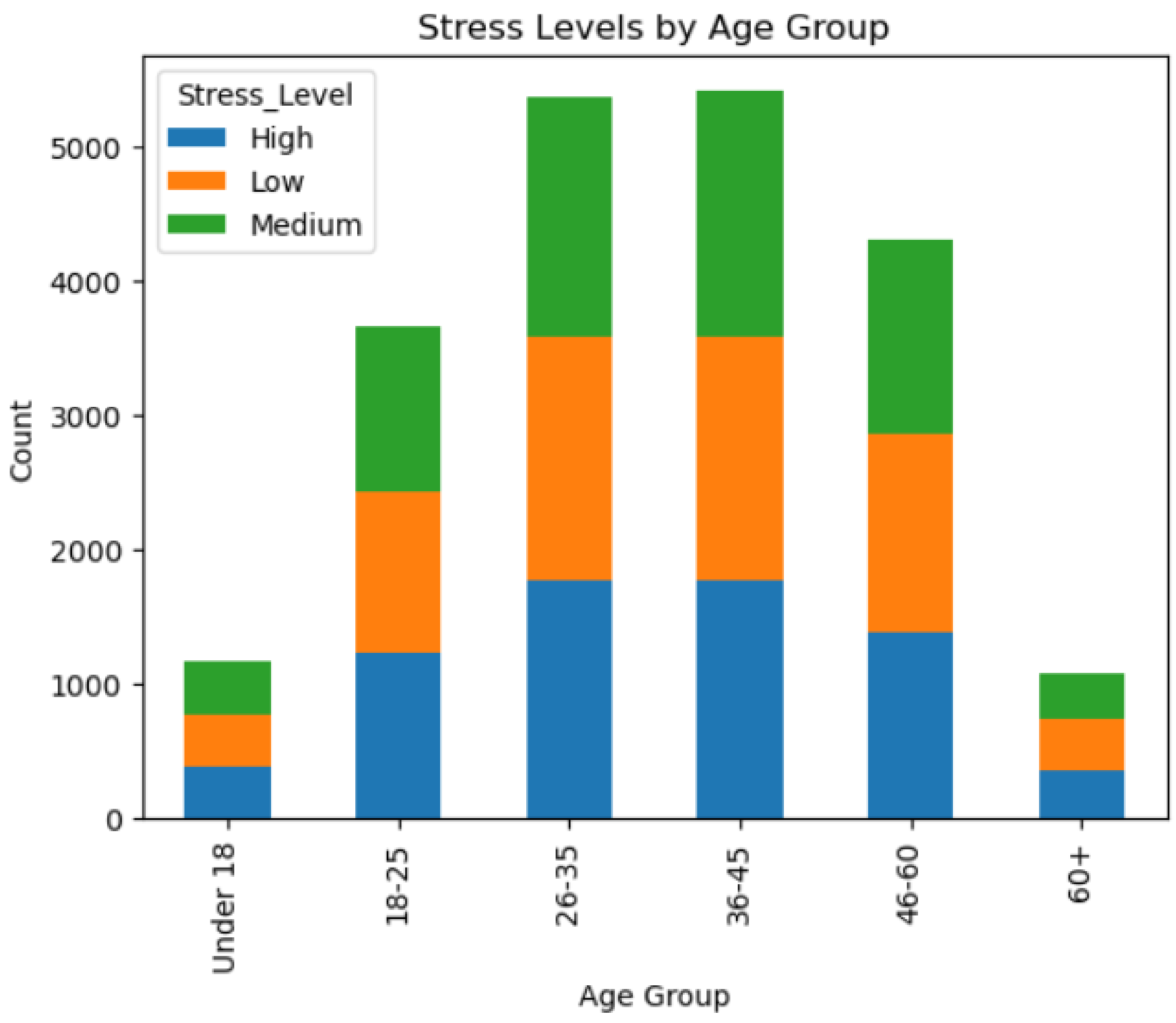
Minimal columns left (7 columns)

- User_ID
- Age
- Social_Media_Usage_Hours
- Screen_Time_Hours
- Stress_Level
- Sleep_Hours
- Physical_Activity_Hours

Research Questions

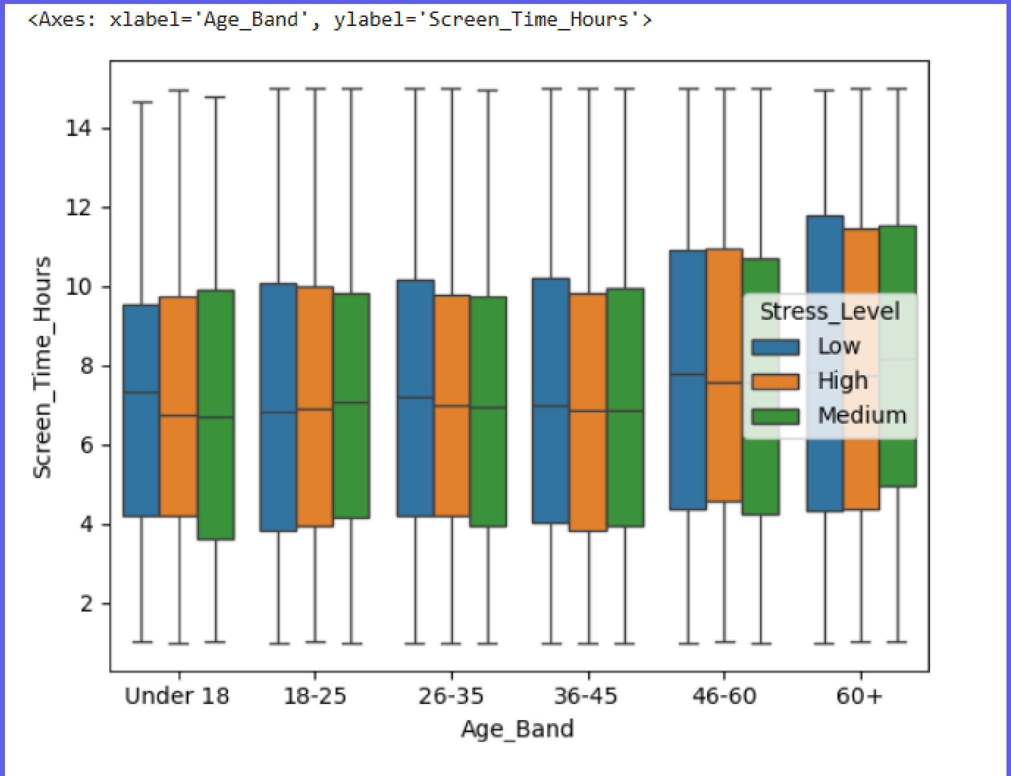
+ How does technology usage affect stress levels and mental health?
+ Does increased screen time correlate with higher stress levels?

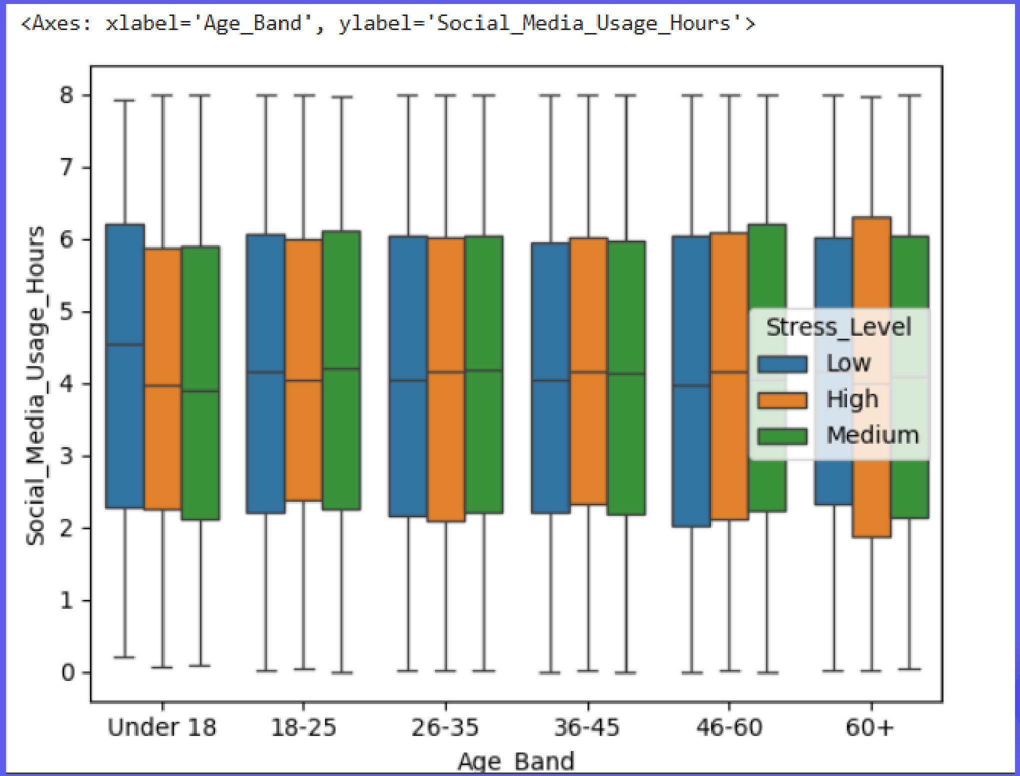
Individuals aged **26–45** represent the largest proportion of the dataset and account for the highest number of reported stress levels. While stress is present across all age groups, the distribution of High, Medium, and Low stress is equally distributed across all ages.

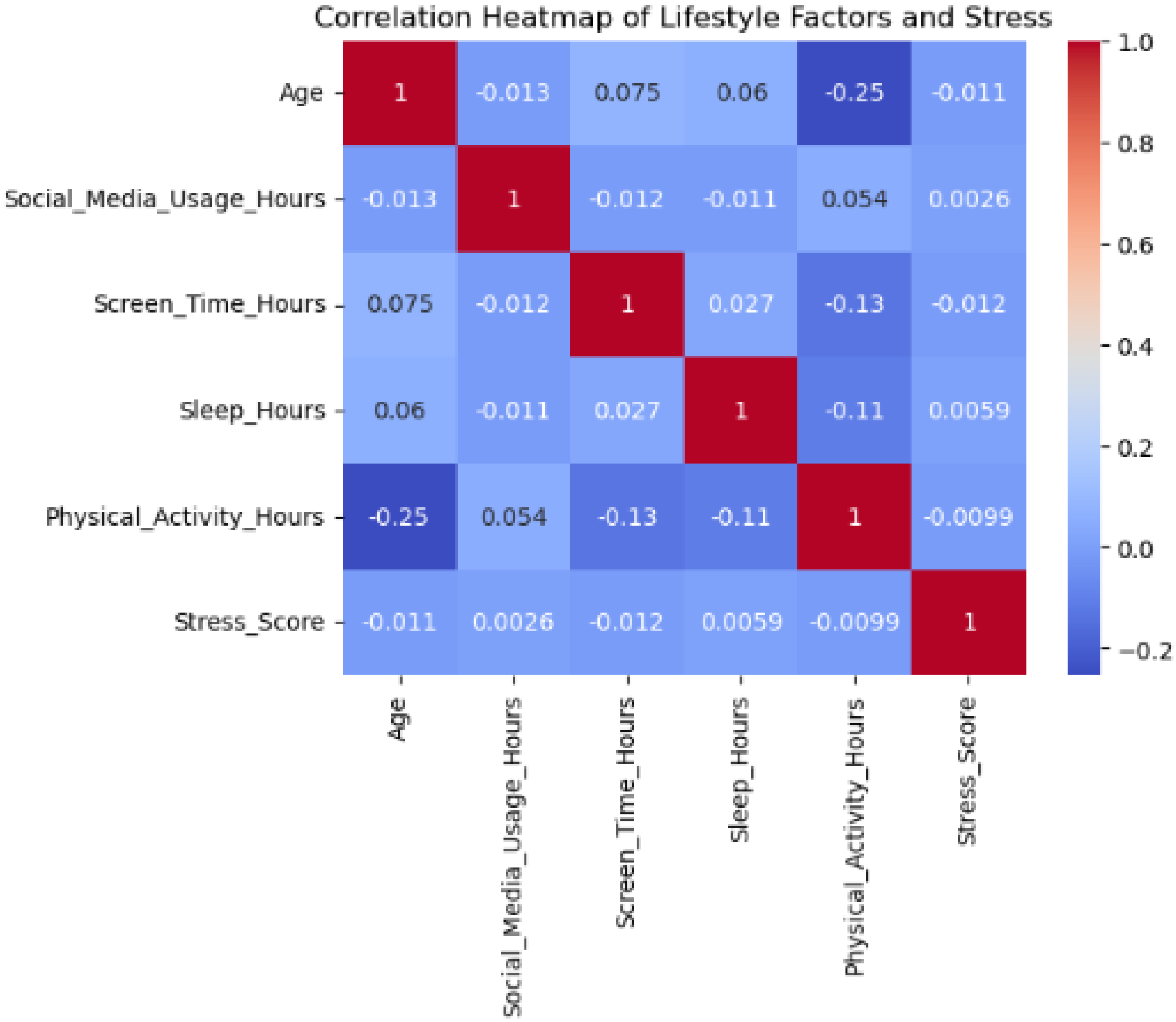


Social Media and Screen Time are similar across all age bands. Surprisingly, screen time increases in older age groups.

Stress levels show substantial overlap across all age groups, indicating that screen time and social media does not predict stress levels.







Definition:
Close to +1 = strong positive relationship
Close to -1 = strong negative relationship
Close to 0 = weak/no relationship

The correlation heatmap indicates that most relationships between stress levels and lifestyle variables are weak. Stress levels do not appear to have a strong linear association with social media usage, screen time, sleep, physical activity, or age in this dataset. The strongest observed relationship was a weak negative correlation between age and physical activity, suggesting that physical activity tends to decrease slightly with age

Summary:

1

This dataset doesn't show any remarkable findings.

2

The dataset became limited after the data cleaning process, as several columns contained missing values and had to be removed.

3

Stress levels show substantial overlap across all age groups, indicating that screen time and social media does not predict stress levels (in this dataset).